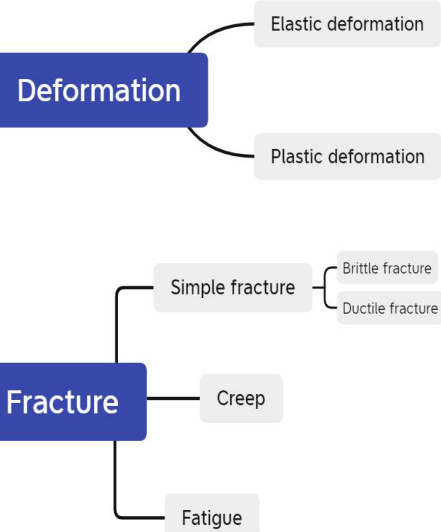


Introduction

To apply the concept of **deformation and failure mechanism** to everyday objects, we designed an experiment to measure the characteristics of **metal spoons, cans and paper clips** in response to stress. This poster will show you the design of experiment and the corresponding experimental results and analysis conclusions of each sample.

Background Information



Objects	Materials	Characteristics
Cans	Aluminum alloy	Brittle Suitable for qualitative manufacturing
Metal spoon	Martinson stainless steel	High temperature resistance Corrosion resistance Toughness
Paper clip	Wire (iron)	Low hardness Strong bending

Material and characteristics of samples

Objective of the project

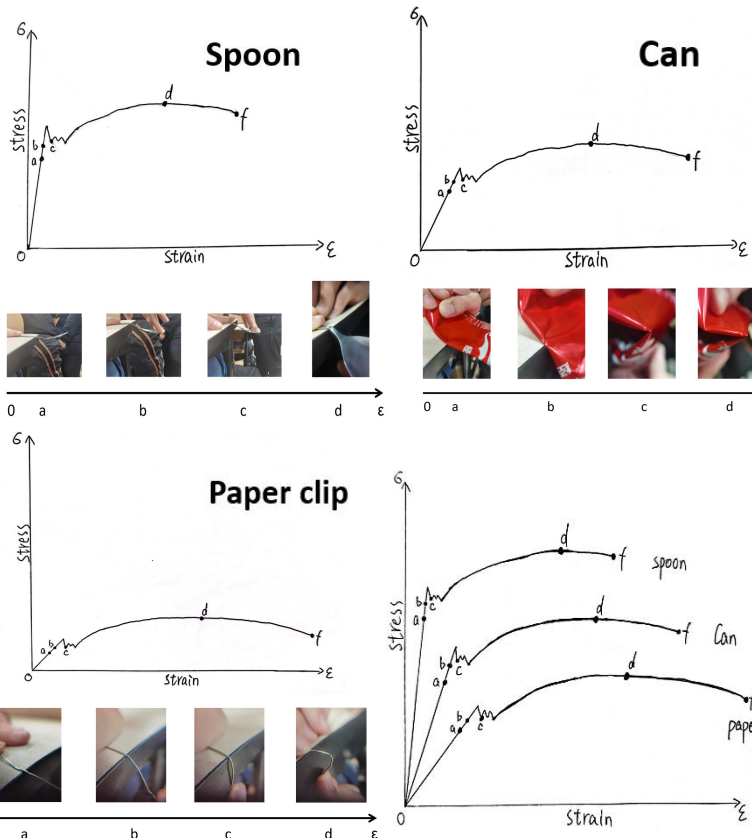
- ✓ Research the deformation and failure in an object
- ✓ Apply deformation and fracture theorems to daily object and explain it in micro structure
- ✓ Develop the ability that cooperate with other members

Methods of experiment

1. Place one side of the material on the table and sticking out the other side. (The cut-off point is the object's center of gravity)
2. Apply **different amounts of force** to the part sticking out of the table
3. Take pictures of the deformation corresponding to the different forces and draw **stress-strain diagram** (4 control groups)

Results and Analysis of findings

Results: Stress-strain diagram of three samples



Analysis:

Similarity: Three samples **elastic deformed** at first (o to b), then **plastic deformed** (c to d) and **fracture** in failure stress (point f).

Differences:

Spoon: **Fatigue failure, high stress and low strain**

Martensitic stainless steel has a **large amount of carbon** which presents a **high hardness in the microstructure**.

Fracture section is in the shape of an **annual ring**, and it can be speculated that **fatigue has occurred**.

Paper clip: **Ductile failure, low stress and high strain**

Iron wire consists arranged as **ordered metal crystals** in microstructure, so they have strong **ductility**. Cracks appear in the wire, when **crack-tip stress (σ_m) exceeds critical stress (σ_c)**, ordered metal crystals stored the energy of deformation and cause ductile failure.

$$\sigma_m = 2\sigma_o \left(\frac{a}{\rho_t} \right)^{1/2} = K_t \sigma_o > \sigma_c = \left(\frac{2E\gamma_s}{\pi a} \right)^{1/2}$$

Can: **Brittle failure, high stress and low strain**

Aluminum alloy consist of many microcrystals, which presents the property of **brittle**. Cracks appear in the cans, when **crack-tip stress (σ_m) exceeds critical stress (σ_c)** and **fracture toughness K_c** crack extend outward and cause brittle failure.

$$\sigma_m > \sigma_c \quad K_c = Y\sigma_c \sqrt{\pi a}$$

Conclusions

We apply deformation and fracture mechanisms concepts to everyday objects. Discussed the three real products from four aspects to explain the process and establish relationships between deformation and failure mechanisms and object chemical composition, micro-structure and manufacturing process. We have successfully combined knowledge with practice.

**Materials Science 1
Deformation and Failure
Reflection of Group work**

1. Describe the focus of the reflection. What was the objective you set out to achieve? Did you achieve this objective?

To apply the concept of deformation and failure mechanism to everyday objects, we designed an experiment to measure the characteristics of metal spoons, cans and paper clips in response to stress. Through the experiment, we achieve the objective we want to set out.

2. Why did the team achieve/ not achieve your objectives? What challenges did you face? How did you overcome these challenges?

The most important reason why we achieve the goal is that we design the experiment process in detail. We think about many possible problems we may face in doing the experiment, so we feel confident when we conduct it. The biggest challenge we face is that the can is hard to break, so we try to flatten the can and then break it crosswise. In this method, we can clearly observe the fracture surface.

3. What have the team learned from this?

We learned that a team is more than the sum of its members' abilities. As a team, the most important thing is to identify our strengths and connect each person's strengths to create a whole that is stronger than any one person. When this is done, the members develop their strengths and make up for each other's weaknesses, thus creating a good team. In addition, team members should frequently discuss problems, brainstorm and share their ideas with others.

4. What changes will you make in the way you work as a team?

We are a very good team and don't need to make wholesale changes, but there are still areas for us to improve. We should increase the number of discussions. At the beginning of the discussion, we only determined the basic idea of making the poster, but did not discuss the details. This led to a lot of problems during making, and we didn't work well together. Finally, we discussed it for many times and achieved our expected result after many modifications.

5. State the contributions from each member of the group to complete the experiment, discussion of results, poster and reflection.

(Example: Student 1: Andy Bushby, designed poster, created graphs, wrote Q4 of reflection)

Student 1: RuiKai Chen, designed poster, conclusions.

Student 2: Runpeng Jiang, methods, results, wrote Q3&Q4 of reflection.

Student 3: Yizhi Zhao, background information, objective of the project, wrote Q1&Q2 of reflection.

Student 4: Yiheng Yang, analysis of findings.